

Rutherford Model

- Rutherford concluded that the majority of the space in an atom is empty. The entire positive charge and mass of an atom is concentrated in a small space known as the nucleus. Electrons revolve around the nucleus in circular orbits.
- The perpendicular distance between the initial velocity vector of the alpha-particle from a central line passing through the center of nucleus when the alpha-particle is far away from the nucleus is known as impact parameter.
- Distance of closest approach: Distance of a point from nucleus at which α -particles are nearest to the centre of nucleus.
- The centripetal force required to make an electron move in a circular orbit is provided by the electrostatic force of attraction between the negatively charged electron and the positively charged nucleus.
- Electrons are bound to the nucleus because the total energy of an electron in orbit is negative.
- Rutherford's atomic model could not explain the stability of the atom. It was also not able to explain the line spectra for various elements and the energies of electrons and their distribution around the nucleus.

Bohr's Model

- According to Bohr's first postulate, every atom consists of a central core called nucleus in which the entire mass and positive charge of the atom is concentrated. Electrons revolve around the nucleus in circular orbits. The centripetal force required for the circular motion of electrons is provided by the electrostatic force of attraction between the negatively charged electrons and the positively charged nucleus.

$$\frac{(Ze)e}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}$$

- According to Bohr's second postulate, an electron can revolve in certain fixed orbits known as stationary orbits. When an electron revolves in a stationary orbit its energy is constant and the angular momentum of the electron is an integral multiple of $h/2\pi$, where h is Planck's constant.

$$mvr = \frac{nh}{2\pi}$$

- According to Bohr's third postulate, when an electron jumps from an inner stationary orbit to an outer stationary orbit

it absorbs energy and when it jumps back from an outer stationary orbit to an inner orbit it radiates energy.

$$E_i - E_f = h\nu = \frac{hc}{\lambda}$$

- Bohr radius:** First orbit of hydrogen atom is called Bohr radius (a_0).
- Ground state:** Lowest state of atom, called the ground state, is the state in which electron revolves in the orbit of smallest radius, the Bohr radius, a_0 .
- Ionization energy:** Minimum energy required to free an electron from the ground state of an atom is called the ionization energy.
- The stationary orbits are not equally spaced. The ratio of radii of the first three stationary orbits is $1^2 : 2^2 : 3^2 = 1 : 4 : 9$.

Spectral lines and Hydrogen Like Atoms

- In hydrogen like atoms,

$$\text{Radius of } n^{\text{th}} \text{ orbit, } r_n = a_0 \left(\frac{n^2}{Z} \right), a_0 = 0.529 \text{ \AA}$$

$$\text{Orbital speed, } v_n = v_0 \left(\frac{Z}{n} \right), v_0 = 219 \times 10^6 \text{ m/s}$$

$$\text{Energy in } n^{\text{th}} \text{ orbit, } E_n = E_0 \left(\frac{Z^2}{n^2} \right), E_0 = -13.6 \text{ eV}$$

- For central forces like in atoms, $TE = -KE = \frac{PE}{2}$
- Wavelength corresponding to spectral lines is given by the Rydberg's formula,

$$\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$R = \text{Rydberg constant} = 1.097 \times 10^7 \text{ m}^{-1}$$
- Total number of possible transitions between any two states is $\frac{n(n+1)}{2}$, where n = difference between the two state numbers.
- The lines for hydrogen are said to be grouped into series, according to the level at which upward jump starts and downward jump ends.

- ❖ When an electron jumps from a higher energy state n to ground state, $n = 1$, the series of spectral lines emitted is called Lyman series. All of them lie in the ultraviolet region.
- ❖ When an electron jumps from a higher energy state n to state 2, the series of spectral lines emitted is called Balmer series. These line in visible region.
- ❖ When an electron jumps from a higher energy state n to state 3, the series of spectral lines emitted is called Paschen series. All of them lie in near infrared region.
- ❖ When an electron jumps from a higher energy state n to state 4 the series of spectral lines emitted is called Brackett series. All of them lie in infrared region.
- ❖ When an electron jumps from a higher energy state n to state 5 the series of spectral lines emitted is called Pfund series. All of them lie in infrared region.

When Effect of motion of Nucleus is Considered,

$$\text{Radius of the orbit, } r_n = (0.529 \text{ \AA}) \frac{n^2}{Z} \cdot \frac{m}{\mu}$$

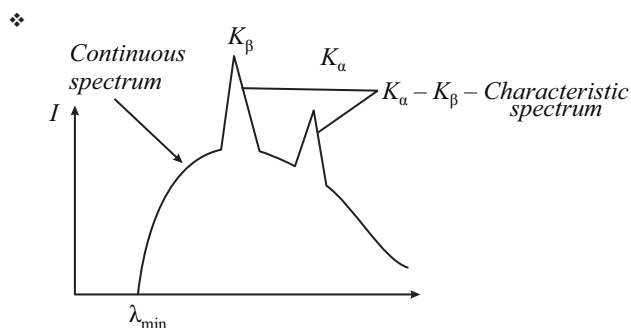
$$\text{Energy of the orbit, } E_n = (-13.6 \text{ eV}) \frac{Z^2}{n^2} \cdot \frac{\mu}{m}$$

$$\text{Here } \mu \text{ is reduced mass, } \mu = \frac{Mm}{(M+m)},$$

Where, M = mass of nucleus, m = mass of charge particle

X-Rays

- ❖ X-rays are produced by bombarding high speed electrons on a target of high atomic weight and high melting point.
- ❖ X-ray are electromagnetic radiations of short wavelength (0.1 Å to 10 Å).
- ❖ Are not affected by electric and magnetic fields.
- ❖ They cause photoelectric emission.



$$\text{Characteristics for continuous X-rays, } eV = h\nu_{\max} = \frac{hc}{\lambda_{\min}}$$

e = electronic charge;

V = accelerating potential

$\nu_{\max}$ = maximum frequency of X-Rays

- ❖ Intensity of X-rays depends on number of electrons hitting the target.

+ Cut off wavelength or minimum wavelength,

$$\lambda_{\min} \cong \frac{12400}{V} \text{ \AA}$$

+ Where V (in volts) is the potential difference applied to the tube

+ Continuous spectrum is due to retardation of electrons.

- ❖ Characteristic X-Rays

$$\text{For } K_{\alpha}, \lambda = \frac{hc}{E_K - E_L} \quad \text{For } K_{\beta}, \lambda = \frac{hc}{E_L - E_M}$$

- ❖ Moseley's Law, $\sqrt{\nu} = a(Z - b)$, here a and b are positive constants for one type of X-rays (independent of Z).